⚡ 20 min Mobile Sprint

For the commute. Read top to bottom. Each block = the highest-value stuff only.

0–3 min Algebra formulas

- $a^2 + b^2 = (a+b)^2 - 2ab$
- $(a-b)^2 = (a+b)^2 - 4ab$
- $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$
- $x^3 + \frac{1}{x^3} = (x + \frac{1}{x})^3 - 3(x + \frac{1}{x})$
- $a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$
- remainder $= f(a)$
- $n(P(A)) = 2^n$, proper $2^n - 1$
- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

3–6 min Log, equations, ratio, series

- $\log_a MN = \log M + \log N$
- $\log M^r = r \log M$
- $\log_a a = 1$, $\log_a 1 = 0$
- base > 0, $\neq 1$
- characteristic = power of 10 in $a \times 10^n$
- never divide by x ($x^2 = 3x \Rightarrow \{0, 3\}$)
- check roots in $\sqrt{}$ and denominators
- যোজন-বিয়োজন $\frac{a+b}{a-b} = \frac{c+d}{c-d}$
- $b^2 = ac$
- AP $t_n = a + (n-1)d$, $S_n = \frac{n}{2}\{2a + (n-1)d\}$
- GP $t_n = ar^{n-1}$
- $\sum n^3 = \{\frac{n(n+1)}{2}\}^2$

6–9 min Trig

- sin: $0, \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}, 1$
- cos reversed
- $\tan 30 = \frac{1}{\sqrt{3}}$, $\tan 60 = \sqrt{3}$
- $\sin^2 + \cos^2 = 1$
- $(\sec + \tan)(\sec - \tan) = 1$
- $\sin(90 - \theta) = \cos\theta$
- height $= d\tan\theta$
- ladder height $= l\sin\theta$
- $30° \to 60°$ walking $a \Rightarrow h = \frac{\sqrt{3}}{2}a$

9–13 min Geometry

- bisectors of B, C $\Rightarrow \angle BOC = 90 + \frac{A}{2}$
- midpoint line $= \frac{1}{2}$ base
- inscribed $= \frac{1}{2}$ central
- semicircle 90°
- cyclic opposite sum 180°
- tangent $\perp$ radius
- 2 tangents from outside are equal
- tangents: sep 4, touch-out 3, cut 2, touch-in 1
- touch: $d = r_1 \pm r_2$
- similar area ratio $=$ side ratio^2
- bisector $\frac{BD}{DC} = \frac{AB}{AC}$
- symmetry: square 4, rhombus 2, equilateral 3, scalene 0, circle infinite
- rotation angle $= \frac{360}{\text{order}}$
- diagonal halves a parallelogram

13–17 min Mensuration + stats

- equilateral $\frac{\sqrt{3}}{4}a^2$
- Heron
- $\frac{1}{2}ab\sin C$
- rhombus $\frac{1}{2}d_1d_2$
- trapezium $\frac{1}{2}(a+b)h$
- arc $\frac{\pi r\theta}{180}$
- cylinder πr^2h
- box diagonal $\sqrt{l^2+b^2+h^2}$
- side $+x\%$ ⇒ area $+(2x+\frac{x^2}{100})\%$
- range = max − min + 1
- $\bar{x}=a+\frac{\sum fu}{n}h$
- Me $=L+(\frac{n}{2}-F_c)\frac{h}{f_m}$
- Mo $=L+\frac{f_1}{f_1+f_2}h$
- ogive y-axis = cumulative frequency
- histogram needs continuous classes

17–20 min Exam moves

- 1. MCQ first, 30 min. Unsure? Eliminate 2, then pick. Don't leave blanks.
  2. CQ: pick one from each group first (Algebra, Geometry, Trig/Mensuration, Statistics), then your strongest.
  3. Theorem: সাধারণ নির্বচন → চিত্র → বিশেষ নির্বচন → অঙ্কন → প্রমাণ.
  4. Every (ক) is 2 easy marks. Write it even if you skip (গ).
  5. Last 10 min: check signs, units, and that you answered what was asked.